

CSE100 Computer Programming

(Common to all branches)

Time: Three hours

Maximum: 100 Marks

Answer all the Questions in SEQUENCE.

PART A

(10 X 2 = 20 Marks)

1. Predict the output:

```
main()
{
    int i=1,j=1;
    for(;;)
    {
        if(i>5) break;
        else j+=i;
        printf("\n%d",j);
        i+=j;
    }
```

2. What is the output for the following code snippet?

```
char ch='A';
while(ch<='F') {
    switch(ch) {
        case 'A':case 'B' : case 'C' : case 'D' : ch++;continue;
        case 'E': case 'F': ch++;
    }
    putchar(ch);
}
```

3. Find the values of a, b, c, d in the following code snippet:

```
main()
{
    int a,b,c,d,i=0,arr[]={11,22,33,0};
    a=arr[++i];    b=arr[i]+3;    c=++arr[a<b];    d=arr[b<c];
    printf("%d\n%d\n%d\n%d",a,b,c,d);
}
```

4. Which of the following prototype declarations are invalid? Why?

- i) int product (int m, int x);
- ii) exchange(int *a, int *b);
- iii) float power(int x,y);
- iv) void fun(void,void);

5. Define an enumeration by the name *trafficlight* with the members RED, YELLOW, GREEN and assign 1, 2 and 3 to the same.

6. Predict the output, (or) predict the error in the following code: Specify the reason

```
main()
{
    int a[]={10,20,30,40,50};
    int j;
    for(j=0;j<2;j++)
    {
        printf("%d ",*a);
        a+=2;
    }
```

7. What would be the output of the following program?

```
#include <stdio.h>
struct point { int x, y;};
void f(struct point *p) {
    p->x =0;
    p->y =0;    }

int main() {
    struct point p = { 2, 3 };
    f(&p);
    printf ("%d%d\n", p.x, .y);}
```

8. Define a structure (using typedef and without using typedef) to represent a rational number (fraction) which contains 2 integer fields one for the numerator and another for the denominator. How do they differ?

9. Consider the declaration:

```
int year=2010; float result=153.675;char word[30]="God is great";
```

Show the output resulting from the following printf statement.

```
printf("YEAR=%6d \n RESULT=%-8.2f \n WORD=%15.3s",year,result,word);
```

10. Predict the output of the following code assuming that the generated code is of 32 bits. Show the steps in the calculation.

```
int main(){
    printf("%x\n", -20 ^ +2);
    return 0; }
```

PART B

(15 X 3 = 45 Marks)

11. Predict the output of the following

```
main()
{ char *s[10]={ "All", "the", " best", " for", " the", " exam"};
  char str[40];
  strcpy(str,s[2]);
  str[strlen(str)]=s[3][1];
  str[5]=s[3][0];
  str[6]=s[0][2]; str[7]='\0';
  strcat(str,"uck");
  printf("%s",str); }
```

12. Write a recursive function to find the sum of the array elements and print the result in the main.

13. A C program contains the following declaration.

```
static int x[8]={ 10, 20, 30, 40, 50, 60, 70, 80};
```

- What will be the output if `printf("%u",x);` is executed?
- What is the value of `*x`?
- What do you mean by `(x+2)`?

14. Evaluate the following and write the final output of s1, s2 and s3.

```
{ int x=10,y=5, z=3,s1,s2,s3;
  s1=!((x==10)||!(y==5)||(z<0));
  s2=!((x<=5)&&(y==10)&&(z<5));
  s3=!((x+y==z)&&!(z>5));}
```

15. Predict the output of the following program

```

int y=1;
call()
{ int p;
  static int s=9;
  for(p=0;p<3;p++)
  {s++; y++; }
  printf("%d %d\n", s, y); }

```

```

main(){
  int x=0, i;
  for(i=-2;i<0;i++)
  { call(); x++; y++; }
  printf("%d %d\n",x, y);}

```

16. Write the output:

```

main()
{ int x=10, y=20, p, q;
  x+= sum(x,y);
  p= sum(x,y);
  q=sum(p,sum(x,2));

```

```

printf("%d %d",p,q); }

```

```

sum(int a, int b)
{ return (a+b); }

```

17. Find the values stored in x, y and z at the end of the program:

```

main(){ int x, y, z, *p, *q;
x=10; y=15; p=&x; q=&z;
*q = *p+y-3; y=y-(*p); *p=*q-z;
z=*p+*q;}

```

18. Predict the output

```

main()
{ // Assume the base address begins at location 1002
  int a[3][4]={1,2,3, 4,5,6, 7,8,9,10,11,12};
  printf("%u %u %u",a[0]+1,*(a[0]+1),*(*(a[0]+1)));}

```

19. Tabulate the scope and lifetime of the storage classes available in C.

20. Predict the output, (or) predict the error in the following code:

```

main()
{ int i,a[5]={2,3,4,5,6};
  change(a);
  for(i=4;i>=0;i--)
  printf("%d ",a[i]);}

```

```

change(int *b)
{ int i;
  for(i=0;i<5;i++)
  *b=*b+1;
  b++;}

```

21. Create 2 structures by the name *name* with the members first name, middle name and last name and *DOB* with the members date, month and the year. Create a union named *employee* with the members name and DOB (which was created earlier).

22. Give the general syntax of malloc and calloc functions. How do they differ?

23. A function *test* accepts pointer to a character and an array of structures of type *cust_rec* and returns a pointer to a structure *cust_rec*. Assume *string* as a character array and *record* as the structure array that is passed to the function. Write the function call and function prototype for function *test*.

24. Define three bit fields, called a, b, c whose widths are 8 bits, 4 bits, 5 bits, respectively. Separate a and b with 2 vacant bits. How will these fields be stored within the computer's memory? (Assume if it is 16-bit memory)
25. Write a program to make use of a macro called *Max* to find the maximum of 2 numbers.

PART C

(5 X 7 = 35 Marks)

26. A book shop uses a personal computer to maintain the inventory of books that are being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the shopkeeper inputs the title and author of the book and the system replies whether it is in the list or not. If it is not, an appropriate message is displayed. If book is in the list, then the system displays the book details and asks for number of copies. If the requested copies are available, the total cost of the books is displayed; otherwise the message "Required copies not in stock" is displayed. (Use appropriate user-defined functions).
27. Write a C program which reads the exponent E (integer) from the user and computes 2^E inside a function *compute* and prints the result. Be sure to handle the case where the exponent is 0 or negative ($2^0 = 1$ and $2^{-E} = 1/2^E$). Note that this requires the result of the computation be type float and not int.
28. Write a program to read an integer array of *n* numbers. Search the key *k* (entered at the run time by the user) in the array. If the key is present delete it from the array and shift the array elements by one position to the deleted place so that there is no empty space in the middle of the array. Repeat the procedure to delete all the occurrences of the key *k*.
29. Define a structure that can describe a *hotel*. It should have members that include the name, address, grade and rooms. Here *rooms* is an inner structure whose members are room charges and number of rooms available. Ask the user to supply room charge(member of the inner structure) and update the grade field of the structure *hotel* based on the conditions below:
- For room charges above 3000, Grade '***' will be provided.
 - For room charges above 2000 and ≤ 3000 , Grade '**' will be provided.
 - For charges above 1000 and ≤ 2000 , Grade '*' will be assigned.
 - For charges below or equal to 1000, display "This is a non-star hotel!"
30. a) Write a program to read a file (that contains alphabets and numbers) and copy the alphabets to the file called *alphabet.txt* and numbers to the file *numbers.txt*. (5 marks)
- b) Write statements using *fseek()* to achieve the following. (2 marks)
- i) To go backward by 9 bytes from the current position.
 - ii) To go forward by 10 bytes from the beginning

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